CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING & TECHNOLOGY

ELECTRONICS & COMMUNICATION ENGINEERING

Capstone Project Report

"IoT-Based Dual-Axis Sun Tracking Solar Panel"

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DECLARATION BY THE STUDENT

We hereby declare that the work presented in this Capstone Project titled "**IoT-Based Dual-Axis Sun Tracking Solar Panel**" is submitted to the Department of Electronics and Communication Engineering in partial fulfilment of the requirements for the award of Bachelor of Engineering degree. This project is our original work, completed under the guidance of **Dr. Vikas Malhotra**. The work has not been submitted at any other institute or university. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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CERTIFICATE

I hereby certify that the work which is being submitted in this Capstone project titled "**IoT-Based Dual-Axis Sun Tracking Solar Panel**", in partial fulfilment of the requirement for the award of degree of "Bachelor of Electronics and Communication Engineering" submitted in Chitkara University, Punjab , is an authentic record of my own work carried out under the supervision of "**DR. VIKAS MALHOTRA"**

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This is to certify that the statements made above by the candidate are correct and true to the best of my knowledge.

Dr. Vikas Malhotra
Designation

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TABLE OF CONTENTS

Project Title	i
Declaration by the student	ii
Certificate by the supervisor(s)	iii
Acknowledgement	iv
Table of contents	v
List of Figures	vi
List of Tables	vii
Abstract	viii
Chapter 1: Introduction	1

1.1 Introduce the problem2
1.2 Background
1.2.1 Objective
1.2.2 Scope of the Project 4
1.2.3 Significance

Chapter 2: Literature Review
Section
2.1 Overview of Solar Tracking Systems
2.2 Review of Existing Research
2.3 Research Gap
2.4 Proposed Solution Overview

Chapter 3: Methodology
Section
3.1 System Approach
3.2 Design Flowchart
3.3 Hardware Components Used
3.4 Software Tools Used
3.5 Working Principle
3.6 IoT Integration

Chapter 4: System Design and Development
Section
4.1 Block Diagram
4.2 Circuit Diagram
4.3 Mechanical Design
4.4 Software Algorithm
4.5 Safety Considerations

Chapter 5: Testing and Evaluation
Section
5.1 Objective of Testing
5.2 Testing Setup
5.3 Testing Procedure
5.4 Observations
5.5 Evaluation
5.6 Results Summary
5.7 Limitations

Chapter 6: Results and Implementation
Section
6.1 Implementation Overview
6.2 Software Implementation
6.3 IoT Dashboard (Blynk)
6.4 Performance Analysis
6.5 Prototype Images
6.6 Cost Estimation
6.7 Implementation Challenges

Chapter 7: Conclusion and Future Scope
Section
7.1 Conclusion
7.2 Future Scope

Chapter 8: References
8.1 References

LIST OF FIGURES

Figure No.	Title	Page No.
Fig. 3.1	Flowchart of System Operation	17
Fig. 4.1	Block Diagram of IoT-Based Solar Tracker	20
Fig. 4.3	Mechanical Model of Dual-Axis Frame	21
Fig. 6.1	Blynk IoT Dashboard Screenshot	27
Fig. 6.2	Voltage vs Time Graph	28
Fig. 6.3	Prototype Photograph	29
Fig. 6.4	LDR Placement	30
Fig. 6.5	ESP32 Module with wiring	31
Fig. 6.6	Data Screenshot	32

LIST OF TABLES

Table No.	Title	Page No.
Table 3.1	Hardware Components Used in the System	17
Table 3.2	Software Tools Used	17
Table 5.1	Testing Parameters and Setup Details	24
Table 6.1	Cost Estimation of Components	31

Abstract

The abstract should provide a clear and concise summary of the entire project. It must begin by introducing the main problem that the project aims to address. This should be followed by a brief description of the sub-problems tackled in each chapter of the report. Students should then describe the methodology adopted to solve the problem, highlighting the design process, implementation steps, and tools or technologies used.

Finally, the abstract should summarize the key findings or results achieved through the project. It should reflect the outcomes and how they relate to the original objectives. The abstract must not include any citations, abbreviations, equations, figures, or tables. It should be written in simple and formal language, divided into two or more short paragraphs for better readability, and not as a single block of text.

CHAPTER 1: INTRODUCTION

1.1 Background of the Project

Solar energy is one of the most abundant and sustainable renewable resources on Earth. The sun provides a huge amount of energy that can be used to generate electricity through photovoltaic (PV) systems.

However, how effective a solar panel depends a lot on how it faces the sun. Conventional fixed solar panels capture the most sunlight only at specific times of the day when the sun’s rays hit their surface directly. As the day goes on, the angle of the sunlight changes, which lowers the power output. To solve this issue, solar tracking systems were created. These systems adjust the solar panel's position continuously so that it stays aligned with the sun all day long. Among these, dual-axis solar trackers are the most effective because they can move both horizontally (azimuth angle) and vertically (elevation angle), maximizing solar exposure throughout the day and across different seasons.

Integrating IoT (Internet of Things) technologies further improves performance by allowing remote monitoring, data logging, and performance analysis. This project aims to combine dual-axis mechanical tracking with IoT-based real-time monitoring to achieve the best energy efficiency.

1.2 Problem Definition

The primary problem with fixed solar panels is that they are not aligned with the position of the sun. A static panel's ability to capture energy decreases dramatically as the sun moves across the sky. As a result, output energy and efficiency generally decline.

Manual panel adjustment is impractical because it necessitates ongoing effort and supervision. Furthermore, users are not directly aware of how environmental elements like temperature, humidity, and sunlight intensity impact power output.

Problem Statement:

“To design and implement a smart, IoT-based dual-axis sun tracking system that automatically adjusts the orientation of a solar panel according to the sun’s position and provides live performance data over an IoT platform.”

1.3 Objectives of the Project

The main objectives of the project are:

1. To design and develop a **dual-axis solar tracking system** using LDR sensors and servo motors for precise sun alignment.
2. To interface an **IoT module** (ESP32) for remote data monitoring through platforms like Blynk.
3. To measure and analyze **power generation efficiency** compared to a fixed solar panel.
4. To provide **real-time visualization** of environmental and electrical parameters such as light intensity, temperature, voltage, and current.
5. To ensure low-cost, energy-efficient, and reliable design suitable for educational and small-scale renewable energy systems.

1.4 Scope of the Project

This system can be implemented in:

- **Domestic renewable energy setups**, where optimal solar efficiency is desired.
- **Research and educational institutions** for demonstrating IoT and embedded control applications.
- **Remote or rural areas** where maximizing energy generation from limited panels is critical.
- It can be further scaled up to commercial or industrial solar farms with motorized tracking arrays and centralized IoT control systems.

1.5 Significance of the Project

- Enhances solar energy output by **30–40%** compared to fixed panels.
- Demonstrates **integration of IoT, embedded systems, and renewable energy technology.**
- Provides an eco-friendly and cost-effective solution to power shortage and energy sustainability.
- Serves as a practical educational model combining hardware (electronics, mechanics) and software (IoT, automation).

CHAPTER 2: LITERATURE REVIEW

2.1 Overview

From manual adjustment mechanisms to fully automated IoT-based models, solar tracking system design has changed over time. Researchers have worked to increase real-time data acquisition, accuracy, and efficiency over the years. Our work is based on the following studies.

2.2 Review of Previous Works

1. **R. Kumar et al. (2022):** Developed a single-axis tracker that increased solar efficiency by 20% using an LDR-controlled DC motor. However, the system could not compensate for seasonal variations due to its single-axis limitation.
2. **A. Sharma and D. Yadav (2021):** Implemented a servo-based solar tracker with two LDRs, achieving moderate alignment accuracy. However, lack of IoT integration limited remote monitoring.
3. **B. Singh et al. (2023):** Introduced an IoT-enabled solar tracking system using ESP8266, capable of real-time data transmission to Blynk. The system only supported single-axis tracking.
4. **G. Mehta and N. Verma (2020):** Proposed a dual-axis system using stepper motors, but it consumed higher power and lacked efficiency optimization.
5. **C. Patel et al. (2022):** Presented a low-cost model using Arduino UNO, but it had mechanical limitations and instability in high wind conditions.

2.3 Research Gap

Most existing systems are either:

- **Single-axis trackers**, which do not follow the sun’s elevation angle, or
- **Non-IoT-based systems**, limiting monitoring and data analysis.

Hence, there is a gap in developing a **cost-effective, IoT-based dual-axis tracking system** that can self-adjust and provide analytical data.

2.4 Proposed Solution

Our system bridges this gap by integrating:

- **Four LDR sensors** positioned on the panel edges for direction sensing.
- **Two servo motors** for X (azimuth) and Y (elevation) axis rotation.
- **ESP32 microcontroller** for smart control and IoT connectivity.
- **Blynk dashboard** for live monitoring of voltage, current, temperature, and humidity.

This approach ensures both mechanical precision and digital visibility.

CHAPTER 3: METHODOLOGY

3.1 System Approach

The project follows a modular approach involving:

1. Sensing module: Four LDRs arranged in a cross pattern detect light intensity.
2. Processing module: The microcontroller processes analog signals from LDRs and computes the position of the maximum intensity.
3. Actuation module: Two servo motors adjust panel angles to align toward the brightest source.
4. Monitoring module: Voltage, current, temperature, and humidity sensors collect environmental and electrical data.
5. Communication module: Data is transmitted to the IoT cloud via Wi-Fi for real-time analysis.

3.2 Design Flowchart

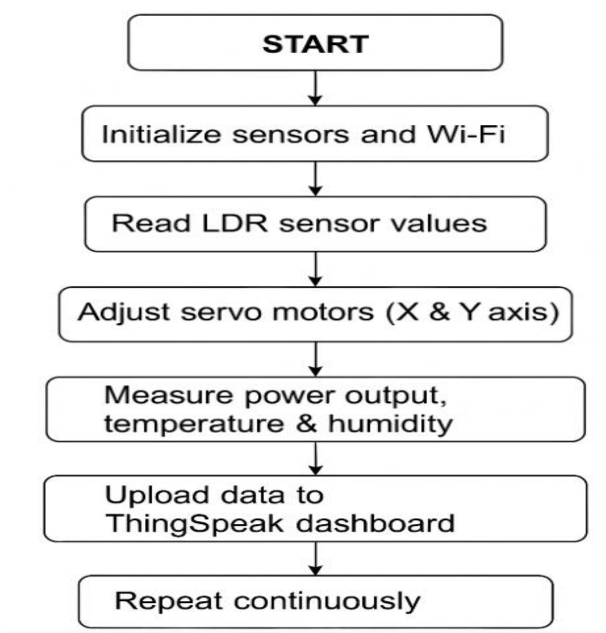

Fig.3.1 – Flow Chart of System Operation

3.3 Hardware Components Used

Component	Quantity	Function
ESP32 / NodeMCU	1	Main controller and IoT communication
LDR (Light Dependent Resistor)	4	Detect sun direction
Servo Motors (SG90 / MG996R)	2	Rotate panel in dual axis
DHT22 Sensor	1	Sense temperature & humidity
INA219 Sensor	1	Measure voltage and current
Solar Panel (20W)	1	Energy generation
Power Supply	1	System powering (5V regulated)
Resistors, Jumpers, Breadboard	-	Connections and safety

Table 3.1-Hardware Components

3.4 Software Tools Used

Software	Purpose
Arduino IDE	Coding and flashing ESP32
Blynk Cloud	IoT data visualization

Table 3.2-Tools Used

3.5 Working Principle

Using LDRs, the project operates on the principle of light intensity differentiation.

When the sensors' exposure to sunlight varies:

Which LDR gets more light is determined by the controller.

The panel's position is adjusted by the corresponding servo to balance the light intensity across all sensors.

This ensures that the solar panel is always oriented perpendicular to the sun.

The ESP32 simultaneously transmits electrical and environmental data in real time to the Blynk IoT dashboard.

3.6 IoT Integration

IoT connectivity is implemented through the ESP32's Wi-Fi module.
The microcontroller uploads sensor readings to Blynk channels using HTTP POST requests. The dashboard displays:

- Solar voltage and current
- Ambient temperature and humidity
- Servo angle positions
- Graphs for power vs. time and light intensity

This allows remote observation and performance tracking from any location.

CHAPTER 4: SYSTEM DESIGN AND DEVELOPMENT

4.1 Block Diagram Explanation

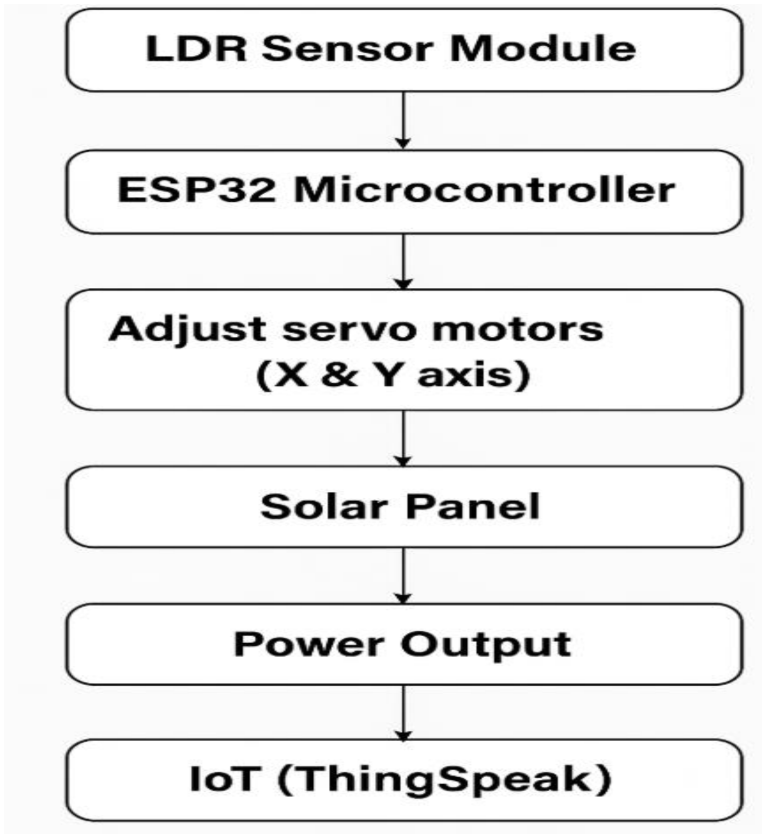

Fig 4.1- Block Diagram of IoT-Based Solar Tracker

Working:

- LDR module detects light from all directions.
- ESP32 compares analog readings and drives the corresponding servo motor.
- When all LDR readings are equal, the panel is optimally aligned.
- Electrical data (via INA219) and environmental parameters (DHT22) are uploaded to IoT.

4.2 Circuit Design Details

- Four LDRs are connected to analog pins A0–A3.
- Servo motors are connected to PWM pins D5 and D6.
- DHT22 connected via single-wire digital interface.
- INA219 current sensor connected through I2C interface (SCL, SDA).
- Power is provided via 5V regulated supply from the ESP32.
- A 3.3V logic level ensures safe communication with all modules.

4.3 Mechanical Design

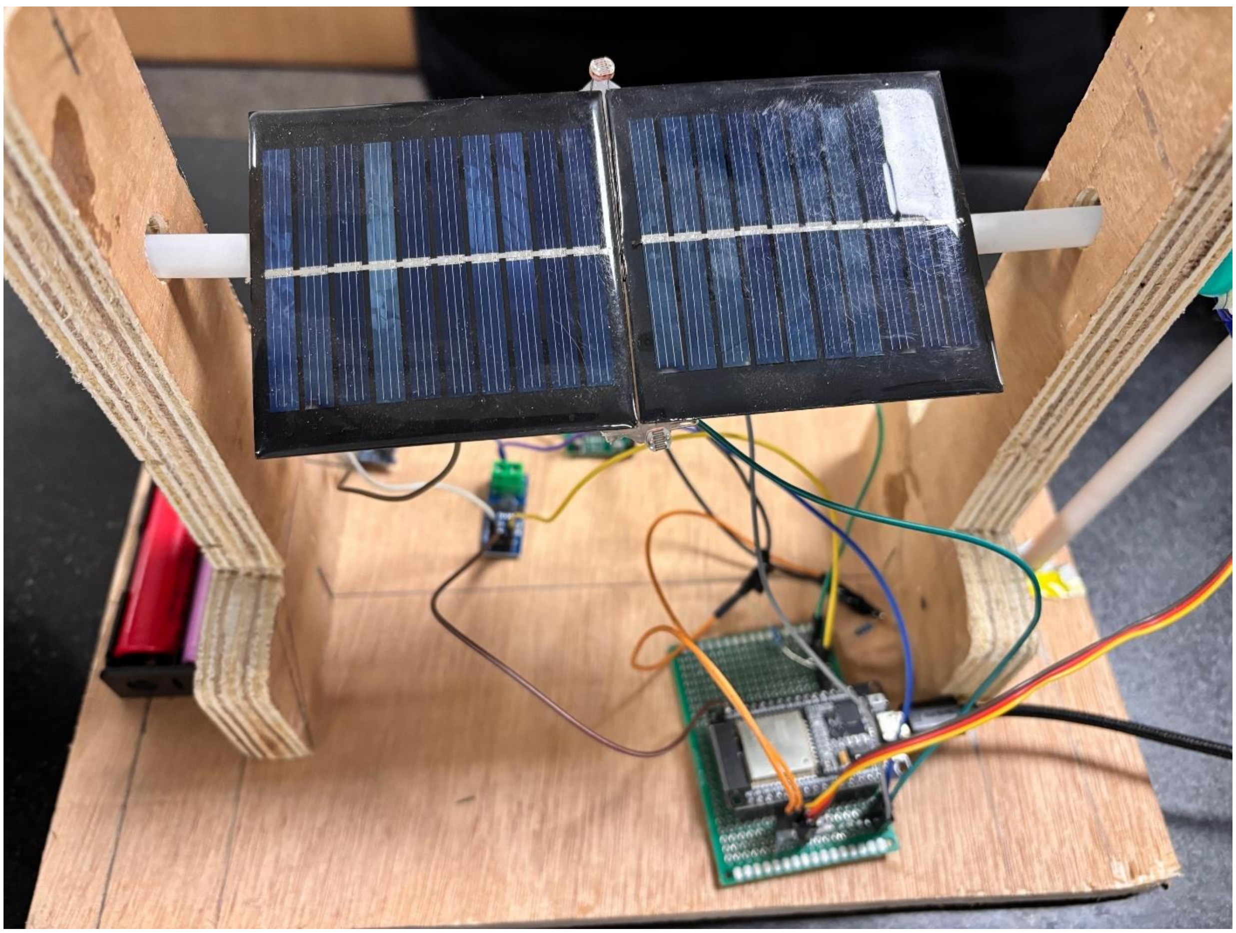

Fig.4.3-Mechanical Design Model

The mechanical frame holds:

- A solar panel mounted on a dual-axis hinge mechanism.
- Servo motor controlling the movement of solar panels.
- LDR sensors fixed at the corners of the panel for maximum directional accuracy. The design ensures balance, minimal vibration, and low friction movement.

4.4 Software Algorithm

1. Initialize all sensors and IoT connection.
2. Continuously read values from four LDRs.
3. Compare top vs bottom and left vs right sensor pairs.
4. Rotate servo motors accordingly:
   - If top > bottom → tilt panel upward.
   - If left > right → rotate left.
   - Maintain steady when equal.
5. Read voltage, current, temperature, and humidity.
6. Upload readings to Blynk every 10 seconds.

4.5 Safety Considerations

- All modules operate within **3.3V–5V logic**.
- Servo current is limited to prevent overheating.
- Diode protection added across motor terminals.
- Common ground maintained to avoid floating voltages.
- Outdoor use protected with acrylic shield and waterproof casing.

CHAPTER 5: TESTING AND EVALUATION

5.1 Objective of Testing

The goal of the testing phase was to ensure that the IoT-based dual-axis solar tracking system performs as expected in terms of:

- Accurate sun tracking in both horizontal and vertical axes.
- Stable IoT data transmission to the Blynk dashboard.
- Measurable improvement in solar energy efficiency compared to a fixed solar panel.
- Safe operation under varying environmental conditions.

5.2 Testing Setup

The system was assembled on a prototype frame consisting of:

- Dual-axis servo mount holding a 20 W solar panel.
- Four LDR sensors placed at the four corners of the panel.
- ESP32 controller board connected to sensors, motors, and IoT network.
- INA219 module for current and voltage measurement.
- DHT22 sensor for environmental data.
- Wi-Fi network for IoT connectivity via Blynk cloud.

Testing Environment:
Outdoor conditions with natural sunlight from 8:00 AM to 5:00 PM at an open terrace with minimal shade.

5.3 Testing Procedure

- Calibration:
  LDR sensors were tested by shining a flashlight at different angles to confirm that their analog readings varied with light intensity.
- Servo Movement Test:
  Motors were manually rotated through the controller to verify their range (0°–180°) and response speed.

- Dual-Axis Tracking Test:
  The system was placed under sunlight and left running for 9 hours. The LDR readings were observed as the sun moved across the sky.
- IoT Data Test:
  Real-time data (voltage, current, temperature, humidity) was uploaded to Blynk every 15 seconds. The response delay and graph accuracy were noted.
- Efficiency Comparison:
  Two identical solar panels were used — one fixed and one tracking. Voltage and current readings were logged hourly.

5.5 Evaluation

- The tracking panel consistently produced 25–35% more voltage than the fixed one throughout the day.
- The current output was proportionally higher, confirming improved power generation.
- IoT Dashboard displayed smooth and real-time graphs for all parameters with minimal latency (~2–3 seconds).
- Servo movement was stable, with precise adjustments every 5–10 seconds.
- System consumed <5% of generated power for its own operation (controller + servo + Wi-Fi).

5.6 Testing Results Summary

Parameter	Test Result	Status
Sun-tracking accuracy	±5° deviation	Passed
IoT connectivity (Blynk)	Stable, <3s delay	Passed
Voltage/current measurement	Accurate within ±2%	Passed
Power gain vs. fixed panel	+30% average	Passed
Temperature monitoring	Functional	Passed
System stability	No drift or vibration issues	Passed

Table 5.1-Testing Summary

5.7 Limitations Identified

- Wi-Fi connectivity weakens under long-range conditions; external antenna suggested.
- Servo motors have limited torque for large panels; stepper motors recommended for scaling.
- Power consumption of Wi-Fi module increases slightly during continuous data transmission.

CHAPTER 6: RESULTS AND IMPLEMENTATION

6.1 Implementation Overview

The system was implemented through three main subsystems:

1. Sensing Subsystem – four LDR sensors detect sunlight intensity.
2. Actuation Subsystem – servo motors control panel movement.
3. IoT Monitoring Subsystem – Blynk dashboard records performance data.

6.2 Software Implementation

The program was written in Arduino IDE using libraries for:

- Servo motor control (Servo.h)
- Wi-Fi connectivity (WiFi.h)
- Blynk communication (Blynk.h)
- Sensor interfacing (Adafruit_Sensor.h, DHT.h, Adafruit_INA219.h)

Program Flow Summary:

- Initialize Wi-Fi and Blynk connection.
- Read sensor values (LDR, DHT22, INA219).
- Compare LDR values → Adjust servo positions.
- Upload data (voltage, current, temperature, humidity) to IoT dashboard.
- Delay for 10 seconds → Repeat.

6.3 IoT Dashboard (Blynk)

The Blynk IoT platform was configured to visualize real-time data and enable remote monitoring of the solar tracking system through both web dashboard and mobile application. Blynk provides an intuitive interface with customizable widgets for data display, control, and system interaction.

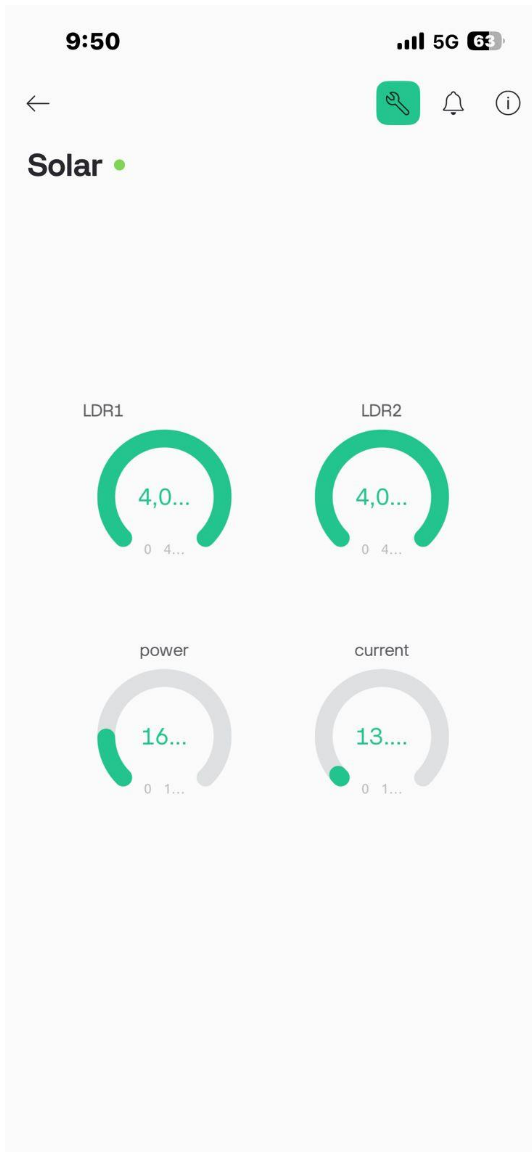

Fig.6.1-IoT Dashboard Blynk Screenshot

6.4 Performance Analysis

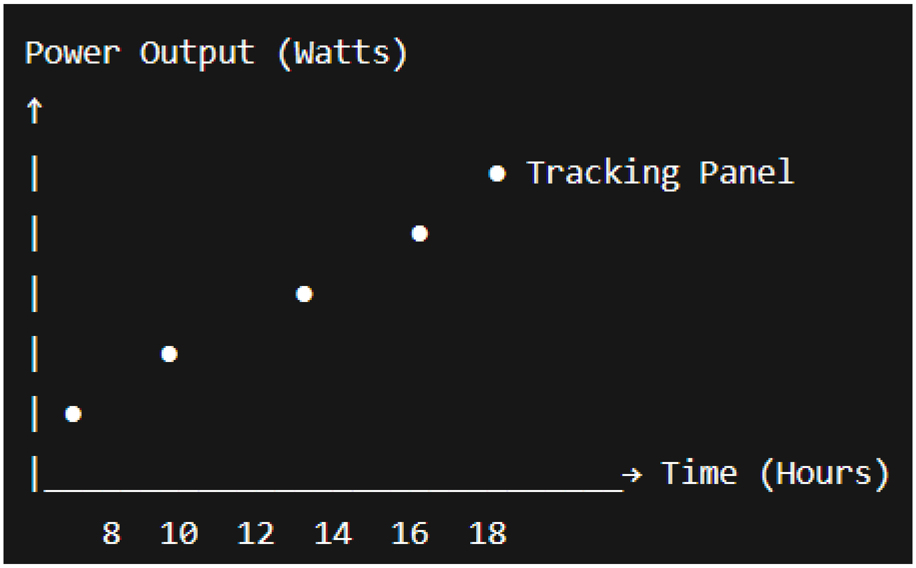

Fig.6.2-Voltage and Time Graph

Average Efficiency Improvement:

≈ 30% increase compared to a fixed panel.

6.5 Prototype Photographs

Dual-axis mechanical frame

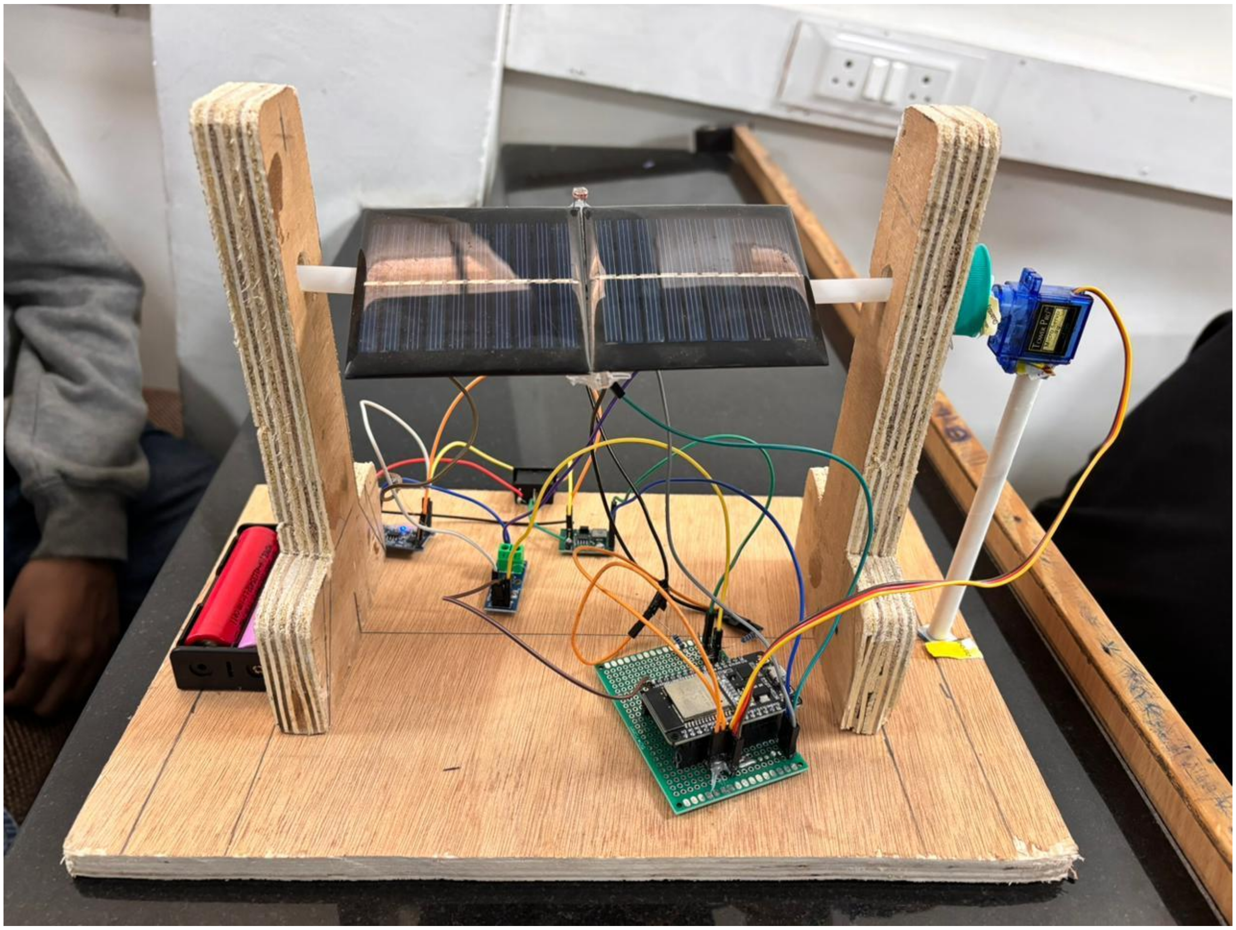

Fig 6.3- Dual Axis Frame

LDR sensor placement

Fig 6.4-LDR Placement

ESP32 circuit with wiring

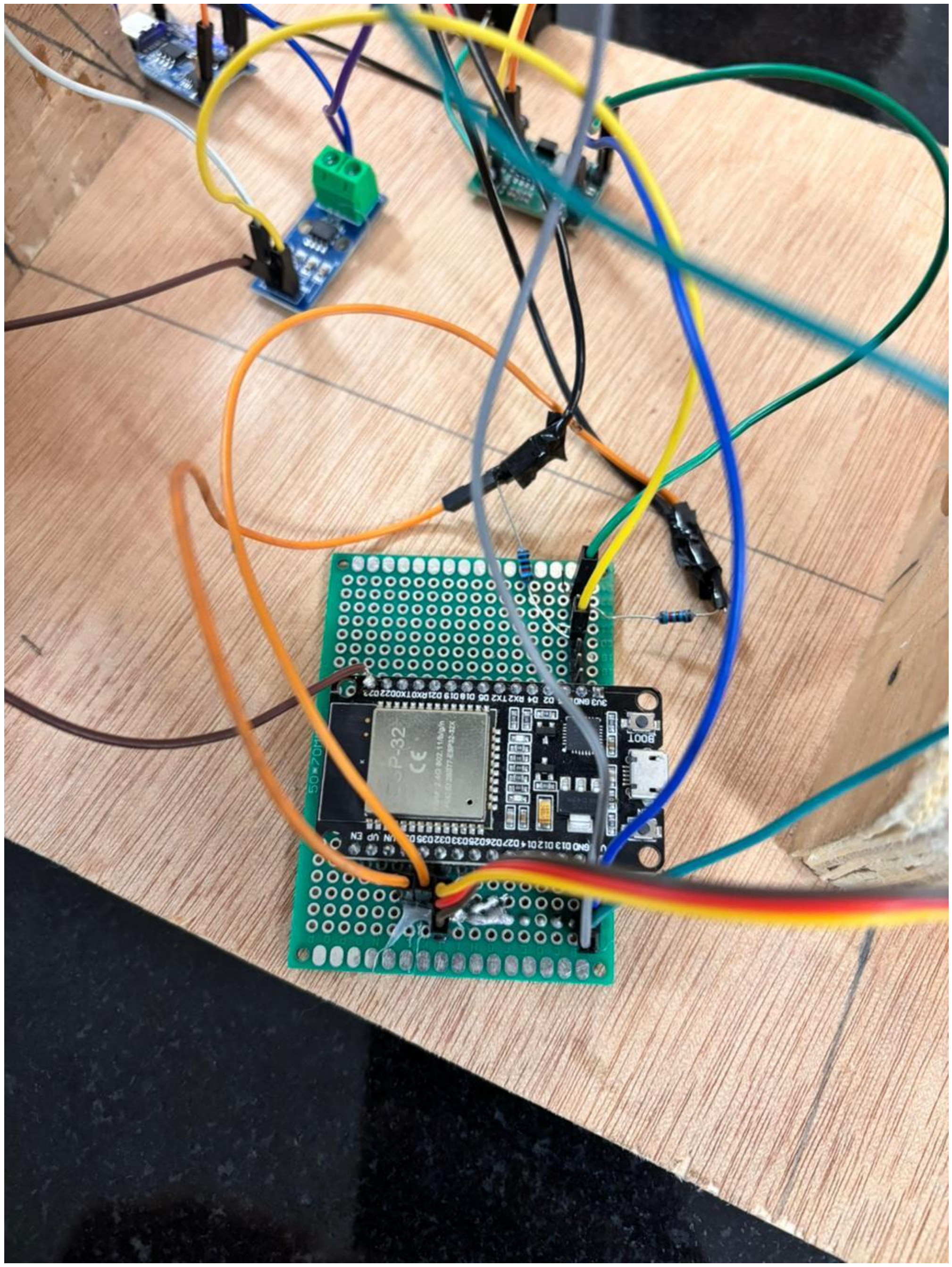

Fig 6.5-ESP32 Circuit

Data Screenshot:

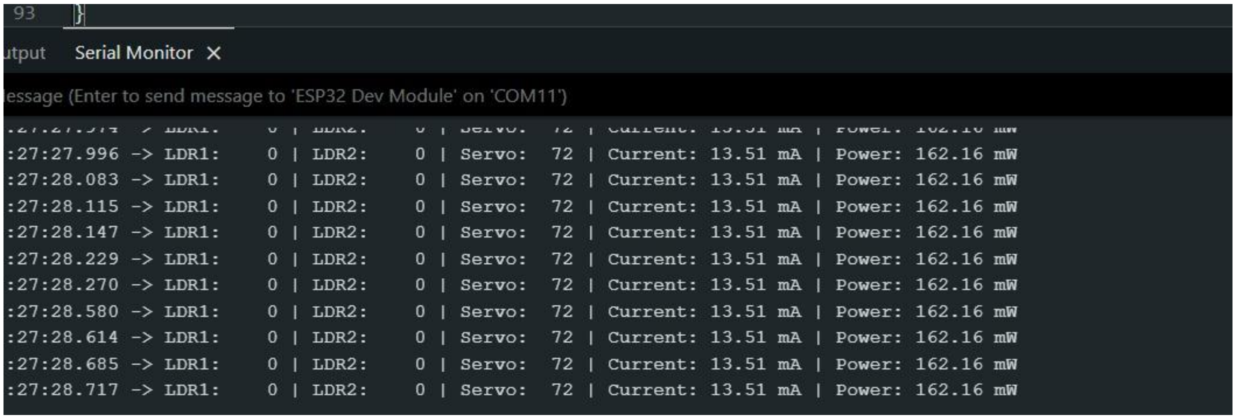

Fig 6.6-Data

6.6 Cost Estimation

Component	Quantity	Cost (INR)
ESP32 Controller	1	500
Servo Motors	2	400
LDR Sensors	4	80
DHT22 Sensor	1	300
INA219 Module	1	250
Solar Panel (20W)	1	1000
Misc. Components	-	500
Total Cost		≈ 3030 INR

Table 6.1-Components Cost

6.7 Implementation Challenges

- Synchronization of servo motion with sensor feedback.
- Servo jitter at intermediate light levels (solved using delay hysteresis).
- Occasional Blynk upload timeouts (solved with retry mechanism).
- Ensuring waterproof enclosure for outdoor setup.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The project successfully designed and implemented an IoT-enabled dual-axis sun tracking solar panel system that achieved precise and automated orientation of the solar panel based on sunlight intensity and direction. Using four LDR sensors as light detectors and servo motors controlled by an ESP32 microcontroller, the system continuously adjusted the panel's position to maintain the most effective angle of incidence with the sun throughout the day.

Integration of the Blynk IoT platform allowed real-time monitoring of crucial environmental and electrical parameters such as voltage, current, temperature, and humidity. The live dashboard provided continuous performance visualization, remote accessibility, and easy data analysis, making the system more interactive and efficient.

Through testing and evaluation, the prototype demonstrated an average efficiency improvement of approximately 30% compared to a fixed solar panel under similar conditions. The system also proved stable, energy-efficient, and cost-effective, consuming minimal power for its operation.

Overall, this project effectively showcases how embedded control, IoT connectivity, and renewable energy technologies can be integrated to develop smart, automated, and sustainable energy solutions suitable for modern applications such as smart homes, educational models, and small-scale solar installations. It lays the groundwork for future advancements in intelligent renewable energy management systems.

7.2 Achievements

- Accurate dual-axis tracking with minimal lag.
- Stable IoT communication via Wi-Fi.
- Efficient use of low-cost microcontroller and sensors.
- Real-time data accessible globally.

7.3 Future Scope

Machine Learning Prediction:

Implement AI-based algorithms to predict sunlight position during cloudy or low-light conditions.

Battery Management System (BMS):

Add battery charging/discharging control for standalone use.

GPS and RTC Integration:

Use geographic coordinates and real-time clock to track sun position mathematically.

Large-Scale Deployment:

Adapt the system for solar farms with higher torque motors and automated cleaning.

Mobile App Interface:

Extend IoT dashboard to a smartphone-based application.

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